



MD RAFIUL ISLAM

ML ENGINEER | COMPUTER VISION • AGENTIC AI

368/2-A, Ahmednagar, Mirpur 1, Dhaka-1216 | rafiulislam1921@gmail.com | +880 1679899117 (WhatsApp) |
www.rafiulislam.me | www.github.com/rafi0020 | www.linkedin.com/in/rafi009

PROFESSIONAL PROFILE

Applied Machine Learning Engineer and researcher who takes AI systems from problem definition and proof of concept through production deployment, validation, and customer handover. Expertise in real-time computer vision, edge inference, multi-camera video analytics, OCR, tracking, pose/face intelligence, and evidence-backed decision systems. Built enterprise AI solutions for banking, manufacturing, logistics, industrial safety, and retail; also developed RAG and workflow-automation platforms using FastAPI, local LLMs, vector retrieval, RBAC, and human-in-the-loop controls. Work across edge AI, AI-enabled security, agentic workflows, APIs, hardware evaluation, technical demos, and customer-facing solution engineering.

EDGE AI

Jetson, ONNX/TensorRT pathways

IMAGE PROCESSING

CNN, Vision Transformer, Image Annotation

VIDEO ANALYTICS

YOLO/OBB, tracking, OCR, pose, face, depth

PRODUCTION SYSTEMS

RTSP, multiprocessing, APIs, SQLite/FAISS

AGENTIC AI

RAG, LLM, LangGraph orchestration

CUSTOMER DELIVERY

POCs, UAT, deployment, troubleshooting

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Bondstein Technologies Ltd. | Feb 2025 – Present

- Delivered end-to-end lifecycle of customer-facing AI solutions across banking, manufacturing, logistics, safety, and retail. From requirements analysis and POC design to field deployment, UAT, monitoring, and support.
- Design low-latency RTSP/GStreamer pipelines using Python, multithreading, multiprocessing, queue backpressure, GPU assignment, auto-reconnect, and restart-safe state management for multi-camera environments.
- Build production perception stacks with PyTorch, Ultralytics YOLO/OBB/segmentation, ByteTrack/DeepSORT/BoT-SORT, PaddleOCR, InsightFace, MediaPipe, Depth Anything, OpenCV, and domain-specific validation.
- Engineered validation-first false-alarm controls using tracking, temporal voters, ROI and homography rules, pose geometry, identity logic, semantic verification, and auditable evidence for high-consequence alerts.
- Deploy workloads on Jetson and GPU servers; prepare ONNX/TensorRT conversion paths, FP16/INT8 deployment plans, benchmark latency/throughput, and tune inference and memory behavior for constrained edge environments.
- Integrate models with REST APIs, SQLite/FAISS, asynchronous senders, Docker/Linux services, evidence snapshots, and audit-ready event queues; document architectures, deployment procedures, and troubleshooting steps.
- Work with client, business-development, project, and field teams to translate operational problems into measurable AI acceptance criteria, technical demonstrations, and reliable production systems.

ENTERPRISE AI/CV SYSTEMS

BRAC CIN AI Surveillance System | BRAC Bank PLC

Architected identity-aware security analytics for an approximately 1,000-camera rollout across ATM, Branch, Sub-branch, and DC/DR sites. Modules include face recognition, unauthorized entry, intrusion, suspicious-object analysis, unusual activity, lying detection, vandalism, camera tampering, expression analysis, and abusive audio. Developed a shared-frame-bus approach to reduce duplicate RTSP decoding and support multi-node scale.

Argus Automata | Unilever Bangladesh Ltd.

Built a modular industrial-safety platform spanning intrusion, calibrated vehicle overspeed, human-vehicle and vehicle-vehicle collision risk, PPE compliance, work-at-height safety, and pose-sequence throwing detection using homography, tracking, relative velocity/TTC, temporal voting, SQLite, and API delivery.

AI Leaf SOP Checker | British American Tobacco Bangladesh Ltd.

Developed a four-line, paired-camera SOP-monitoring system for bale counting and cut/uncut classification, layer/moisture verification, hand-on-bale monitoring, and barcode-stage analysis. Implemented YOLO segmentation/OBB models, seam handling, per-bale state, persistent counts, and asynchronous evidence delivery.

Track My Container | KDS Logistics Ltd.

Deployed an edge AI + IoT container-identification pipeline using fisheye correction, YOLO OBB, vertical/horizontal text reconstruction, PaddleOCR, ISO 6346 validation, and prefix correction. Integrated Jetson Orin Nano with ESP32, GNSS, ultrasonic/BME280 sensing, SIM7600 telemetry, UART/Bluetooth, and backend APIs.

Showroom AI | Runner Automobiles Ltd.

Combined customer/employee identification, repeat-customer capture, footfall, bike entry/exit, product-model recognition, customer-product interaction, service-delay analytics, and shutter monitoring. Used YOLO, BoT-SORT, InsightFace, configurable zones/lines, SQLite state, and decoupled uploaders.

Smart Parking with Bangla ANPR | New Asia Group Ltd.

Engineered vehicle-centric plate sessions with YOLO OBB, ByteTrack, Bengali OCR, Bangladesh plate-format validation, confidence voting, entry/exit deduplication, and API integration; mapped identity events into authorization, bay allocation, occupancy, fee, payment, and barrier workflows.

Mobil Product Inspector | Mobil Bangladesh Ltd.

Built a bottle-label and product-identification application using custom YOLO OBB models, rectified multi-view crops, RapidOCR, barcode/HR-code extraction, confidence-aware l1 correction, catalog matching, mobile browser camera controls, SQLite evidence logs, and reporting.

Staircase Safety Monitoring | Unilever Bangladesh Ltd.

Developed a pose-based safety system using YOLO person detection and MediaPipe Pose to assess handrail compliance and hand-near-ear phone behavior. Added camera-specific rail polygons, temporal state stabilization, annotated evidence, RTSP recovery, and retrievable API delivery.

AGENTIC AI & PLATFORM ENGINEERING

HR Assist | Internal HR Platform

Built a FastAPI/SQLite AI-assisted HR service desk with hybrid BM25 + MiniLM retrieval, OCR-indexed policies, local Ollama generation, Bengali/Banglish normalization, citation-backed answers, dynamic request forms, deterministic approvals, RBAC, segregation of duties, fulfillment queues, internal mail, SLA analytics, and human-reviewed AI drafting.

CV Agent Ecosystem | Personal Project

Created a LangGraph multi-agent CV engineering platform with specialist agents for product, architecture, data, MLOps, code generation, review, testing, documentation, and SRE. Implemented typed shared state, human approval gates, reusable engineering knowledge, model routing, and workspace generation.

VisionGuard | Personal Project

Developed a modular multi-camera analytics control plane using Bun/TypeScript, SolidJS, SQLite, H.264/WebSocket streaming, persistent Python ML workers, remote-engine support, dynamic model/task management, evidence analytics, ROI configuration, VLM agents, and worker health controls.

Unified AI OS | Personal Project

Built a personal multi-model research workspace that routes tasks through local Ollama, coordinates Claude, Gemini, ChatGPT, and Perplexity through parallel Playwright workers, streams progress over SSE, and synthesizes results in a React/FastAPI interface.

Crowd/Mob Detection | Personal Project

Combined a custom YOLO11 person/suspect/victim/weapon model with DeepSORT, Depth Anything V2, depth-aware proximity graphs, Gaussian density heatmaps, temporal smoothing, and weapon-proximity suspect pinning for body-camera or RTSP footage.

RESEARCH EXPERIENCE

Research Assistant | 4IR Research Cell, Daffodil International University | Concurrent Research Appointment

- Conduct research in computational healthcare and trustworthy AI, centered on Bengali voice-based psychological stability assessment using log-mel spectrograms, CNNs, Vision Transformers, and hybrid architectures.
- Designed subject-independent evaluation, data-augmentation, and transfer-learning experiments under limited data; worked with ethically collected voice data from mental-health institutions in Bangladesh.
- Developed research pipelines involving DenseNet–ViT–GRU variants, noise-robust learning, CLAHE enhancement, ensembles, and XAI methods including Grad-CAM, LIME, and SHAP.
- Translate research methods into deployable systems through temporal reasoning, evidence generation, reliability controls, and hardware-aware inference.

PUBLICATIONS

Mental Health Diagnosis From Voice Data Using Convolutional Neural Networks and Vision Transformers

First author • Journal of Voice • Accepted 2024 • DOI: 10.1016/j.jvoice.2024.10.010 • Hybrid CNN–ViT on Bengali voice spectrograms; 91% accuracy and ROC-AUC 0.97.

A Novel Transfer Learning Approach for Mental Stability Classification from Voice Signal

First author • arXiv:2601.16793 • Subject-independent split and augmentation-to-raw fine-tuning; DenseNet121 achieved 94% accuracy and AUC 0.99.

Using Computer Vision for Skin Disease Diagnosis in Bangladesh

First author • arXiv:2501.18161 • Compact DCNN benchmarked against AlexNet, ResNet, VGG16, DenseNet, and MobileNet on HAM10000.

EDUCATION

B.Sc. in Computer Science & Engineering

2020 – 2024

Daffodil International University

CGPA 3.52 / 4.00

Thesis: AI-based psychological stability detection from Bengali voice analysis — hybrid CNN–ViT on spectrogram representations using ethically collected clinical data.

Higher Secondary Certificate (HSC) — BAF Shaheen College Dhaka

GPA 4.45 / 5.00 (2018)

Secondary School Certificate (SSC) — Tamirul Millat Kamil Madrasha

GPA 5.00 / 5.00 (2016)

TECHNICAL SKILLS

AI/ML: Python, PyTorch, TensorFlow/Keras, scikit-learn, CNNs, Vision Transformers, transfer learning, explainable AI

Computer Vision: Ultralytics YOLO v11/v26/YOLOE, OBB, detection, segmentation, tracking, pose, face recognition, monocular depth, homography, ROI logic

OCR & Identity: PaddleOCR, Tesseract OCR, DeepSeek OCR, ISO 6346 validation, InsightFace, FAISS

Edge & Optimization: Jetson Orin Nano, ONNX/TensorRT pathways, CUDA/GPU inference, FP16/INT8 planning, low-latency pipelines, model conversion, benchmarking

Agentic AI & Retrieval: LangGraph, BM25/vector RAG, RRF, embeddings, LLM, vLLM

Video Systems: Multi-stream RTSP, CCTV/IP cameras, shared-memory frame bus, multi-GPU scheduling, NVDEC, GStreamer

Backend & MLOps: FastAPI, REST, Docker, systemd, SQLite/WAL, Git/GitLab, TensorBoard

IoT & Systems: Linux, multiprocessing, ESP32/UART, queues and restart-safe services

CERTIFICATIONS

- IBM Data Science: What is Data Science? and Tools for Data Science (Coursera)
- Generative AI Productivity Skills - Microsoft and LinkedIn | Brain-Computer Interface (BCI) - Pantech.AI Academy

LANGUAGES

Bengali: Native | English: Professional working proficiency (IELTS Academic 6.5)

REFERENCE

Md. Ashfaqul Haque — Sr. Software Engineer (AI & ML), Bondstein Technologies Ltd. — ahq.john@gmail.com — +8801792895969